

بسم الله الرحمن الرحيم

① Poynting Vector: $\vec{S} = \vec{E} \times \vec{H}$
 $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$

② Average S : $S_{avg} = \frac{\vec{E} \cdot \vec{H}}{2}$
 $S_{avg} = \frac{\vec{E} \cdot \vec{B}}{2\mu_0} \xrightarrow{B = \frac{E}{c}} S_{avg} = \frac{E^2}{2\mu_0 c} \text{ or } S_{avg} = \frac{B^2 c}{2\mu_0}$

③ Intensity: $I = \frac{\text{Power}}{\text{Area}} \rightarrow I = S_{avg}$

$I = \frac{E^2}{2\mu_0 c} \text{ or } I = \frac{P}{A}$

④ Total energy = $\frac{P \times L}{c}$

$$\frac{P \times L}{c} = \frac{W \times \cancel{m}}{\frac{m}{s}} \rightarrow \frac{J}{\frac{1}{s}} = J$$

بمساحة الدائرة

⑥ 15 mW laser emits a beam in circular area of diameter $d = 2 \text{ mm}$ in free space

$$r = \frac{D}{2} = \frac{2 \text{ mm}}{2} = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$A = \pi r^2 = \pi \times (1 \times 10^{-3})^2 = \pi \times 10^{-6} \text{ m}^2$$

a) The electric field amplitude is:

$$\frac{P}{A} = \frac{E^2}{2\mu_0 c} \rightarrow E = \sqrt{\frac{P}{A} \times 2\mu_0 c} = \sqrt{\frac{15 \times 10^{-3}}{\pi \times 10^{-6}} \times 240\pi} = 1897.4 \text{ V/m}$$

b) Total energy:
in (1 m) :

$$\text{Energy} = \frac{P \times L}{c} = \frac{15 \times 10^{-3} \times 1}{3 \times 10^8} = 5 \times 10^{-11} \text{ J}$$

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1*

^P 15 mW source emits a beam in ^{in free space} spherical area of radius of 2 mm

a) The electric field amplitude is.

b) Total energy:
in 1m:
L

Problems

Q₁ in free space

$$\vec{E} = 80\hat{i} + 32\hat{j} - 64\hat{k}$$

$$\vec{B} = 0.2\hat{i} + 0.08\hat{j} + 0.29\hat{k}$$

(a) Perpendicular or not: $\vec{E} \cdot \vec{B} = 80 \times 0.2 + 32 \times 0.08 - 64 \times 0.29 = 0 \rightarrow \vec{E} \perp \vec{B}$

(b) Resulting vector $\vec{S}$: $\vec{E} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 80 & 32 & -64 \\ 0.2 & 0.08 & 0.29 \end{vmatrix} = \hat{i}(9.28 + 5.12) - \hat{j}(23.2 + 12.8) + \hat{k}(6.4 - 64)$
 $= 14.4\hat{i} - 36\hat{j}$
 $|\vec{E} \times \vec{B}| = \sqrt{14.4^2 + 36^2} = 38.77$

$$S = \frac{1}{\mu_0} |\vec{E} \times \vec{B}| = \frac{38.77}{4\pi \times 10^{-7}} = 3.09 \times 10^7 \text{ W/m}^2$$

2 *

in free space

$$\vec{F} = 80\hat{i} + 32\hat{j} - 64\hat{k}$$

$$\vec{H} = 0.2\hat{i} + 0.08\hat{j} + 0.29\hat{k}$$

(a) Perpendicular or not:

(b) $\vec{E}$

Problems

بمسألة العقل البشري

Problems

$$c \rightarrow v = \frac{c}{9}$$

$$\mu_0 \rightarrow \mu = (\mu_r \mu_0)$$

$$\epsilon_0 \rightarrow \epsilon = (\epsilon_r \epsilon_0)$$

$$v = \frac{3 \times 10^8}{\sqrt{\mu_r \epsilon_r}}$$

$$c = \frac{1}{\sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}} = \frac{1}{\sqrt{\mu_r \epsilon_r} * \sqrt{\mu_0 \epsilon_0}} \Rightarrow v = \frac{3 \times 10^8}{\sqrt{\mu_r \epsilon_r}}$$

$$\sqrt{\mu_r \epsilon_r} = \left(\frac{c}{v} \right) = 9$$

$$\mu_r \epsilon_r = 4$$

مسألة رقم 3

Q₃ Dish antenna having Diameter of 20m receives a signal of $E_{\text{max}} = 0.2 \text{ V/m}$

$$r = \frac{D}{2} = \frac{20}{2} = 10 \text{ m}$$

$$A = \pi r^2 = \pi \times 10^2 = 100\pi$$

(a) Amplitude of magnetic field: $B = \frac{E}{c} = \frac{0.2 \times 10^{-6}}{3 \times 10^8} = 6.7 \times 10^{-16} \text{ T}$

(b) Intensity of radiation: $I = \frac{E^2}{2\mu_0 c} = \frac{(0.2 \times 10^{-6})^2}{240\pi} = 5.31 \times 10^{-17} \text{ W/m}^2$

(c) Power: $I = \frac{P}{A} \rightarrow P = I \times A = 5.31 \times 10^{-17} \times 100\pi = 1.67 \times 10^{-14} \text{ W}$

(d) Force: $F = \frac{P}{c} = \frac{1.67 \times 10^{-14}}{3 \times 10^8} = 5.57 \times 10^{-23} \text{ N}$

$$\text{Energy} = \left(\frac{F \cdot d}{c} \right) \times L$$

$$\text{Energy} = F \times L$$

$$\text{Energy} = \left(\frac{F \cdot d}{c} \right) L$$

بسم الله الرحمن الرحيم

Q3

Dish antenna having Diameter of 20m receives a signal of $E = 0.2 \text{ mV/m}$

$$r = \frac{D}{2} = \frac{20}{2} = 10\text{m}$$

$$A = \pi r^2 = \pi \times 10^2 = 100\pi$$

$$\text{Energy} = \left(\frac{P}{c} \right) \times L$$

$$\text{Energy} = F \times L$$

$$\text{Energy} = \left(\frac{F \times d}{c} \right) L$$

(a) Amplitude of magnetic field: $B = \frac{E}{c} = \frac{0.2 \times 10^{-6}}{3 \times 10^8} = 6.7 \times 10^{-16} \text{ T}$

(b) Intensity of radiation: $I = \frac{E^2}{2\mu_0 c} = \frac{(0.2 \times 10^{-6})^2}{240\pi} = 5.31 \times 10^{-17} \text{ W/m}^2$

(c) Power: $I = \frac{P}{A} \rightarrow P = I \times A = 5.31 \times 10^{-17} \times 100\pi = 1.67 \times 10^{-14} \text{ W}$

(d) Force: $F = \frac{P}{c} = \frac{1.67 \times 10^{-14}}{3 \times 10^8} = 5.57 \times 10^{-23} \text{ N}$

3*

Dish antenna having radius of 20m receives a signal of $B_{\text{max}} = 2 \text{ kV/m}$

(a) Amplitude of electric field:

(b) Intensity of radiation:

(c) Power:

(d) Force:

(e) Energy in 3m: